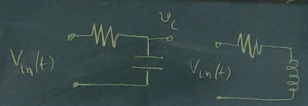
951

Goal. AC circuits ← headed today



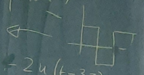
So far, $v_{in}(t) = V u(t)$ or $V \delta(t)$

(see in s-dom) linear

$\frac{d}{dt} u(t)$

linear comb of $u(t)$ } $u(t) - 2u(t-T) + 2u(t-2T) - 2u(t-3T) + \ldots$

→ V pulse(t)

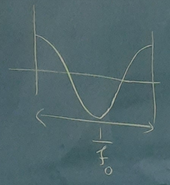


AC circuits

$$V_{in}(t) = V \cos(\omega_0 t + \phi)$$

$$\omega_0 = 2\pi f_0$$

$\phi = 0$



RC

time domain KVL (diff eqn)

$$RC \frac{dv_c}{dt} + v_c(t) = v_{in}(t), \quad v_c(0) = 0$$

So far, solved this for $v_{in}(t) = u(t)$ / lin cal

RC (s-domain)

$$I(s) = \frac{V_{in}(s)}{R + \frac{1}{sC}} \Rightarrow V_C(s) = \frac{1}{sC} I(s) = \frac{V_{in}(s)}{sRC+1}$$

$$v_{in}(t) = V u(t) \Rightarrow V_c(s) = \frac{V}{s} \cdot \frac{1}{sRC+1}$$

$\downarrow$ partial fraction $\Rightarrow V\left(1 - e^{-t/RC}\right)$

Let's consider $V_{in}(t) = e^{at} u(t)$

$$\int_0^\infty e^{at} e^{-st} dt$$

if $s > a$

$$V_c(s) = \underbrace{V_{in}(s)}_{\downarrow \frac{1}{s-a}} \frac{1}{sRC+1}$$

if $\underline{s > a}$

$$\frac{1}{\text{den is deg 2 with roots } a \ \& -\frac{1}{RC}}$$

Part sln $= \frac{A}{s-a} + \frac{B}{s+\frac{1}{RC}}$

$$\left[\frac{1}{\prod_{i=1}^{K} (s - r_i)} = \sum_K \frac{A_K}{s - r_K} \right.$$

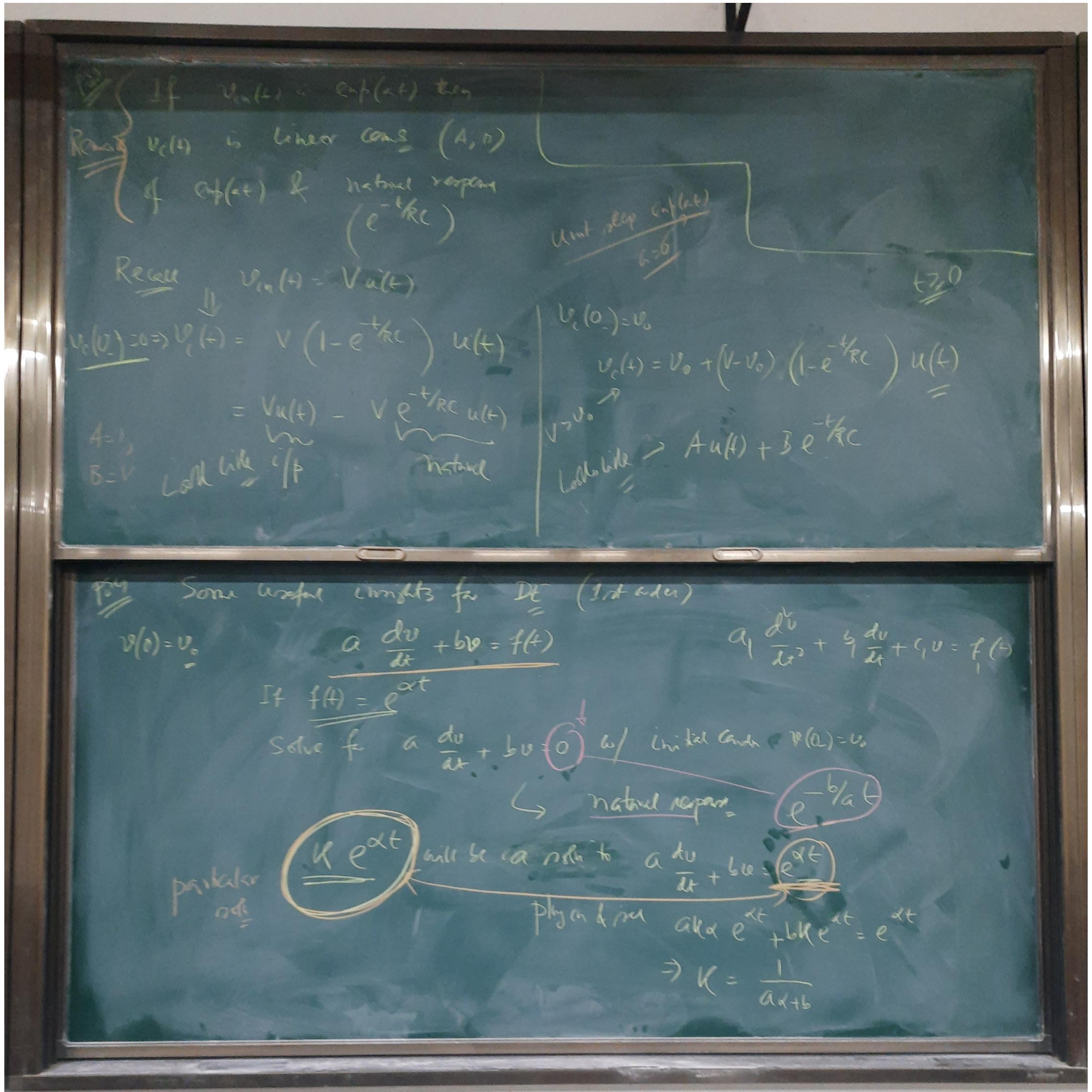
$$\mathcal{L}^{-1} \Rightarrow \boxed{A e^{at} + B e^{-t/RC}}$$

i/p source

init

natural

initial cond (A+B=0)



P3

If $v_{in}(t)$ is exp(αt) then

Remark: $v_c(t)$ is linear comb (A, B) of exp(αt) & natural response $\left(e^{-t/RC}\right)$

Recall $v_{in}(t) = V u(t)$

$\Downarrow$

$$v_c(0_-)=0 \Rightarrow v_c(t) = V\left(1-e^{-t/RC}\right)u(t)$$

$$= \underbrace{Vu(t)}_{} - V\underbrace{e^{-t/RC}u(t)}_{}$$

A=1, B=V

Look like i/p natural

$v_c(0_-)=v_0$

$$v_c(t) = v_0 + (V-v_0)\left(1-e^{-t/RC}\right)u(t)$$

$V \to v_0$

Looks like $\rightarrow Au(t) + Be^{-t/RC}$

P4 Some useful insights for DE (1st order)

$v(0)=v_0$

$$a\frac{dv}{dt}+bv=f(t)$$

$$a_1\frac{d^2v}{dt^2}+b_1\frac{dv}{dt}+c_1v=f_1(t)$$

If $f(t)=e^{\alpha t}$

Solve for $a\frac{dv}{dt}+bv=0$ w/ initial cond $v(0_-)=v_0$

↳ natural response $= e^{-\frac{b}{a}t}$

particular soln: $Ke^{\alpha t}$ will be a soln to $a\frac{dv}{dt}+bv=e^{\alpha t}$

plug in & see $aK\alpha e^{\alpha t}+bKe^{\alpha t}=e^{\alpha t}$

$$\Rightarrow K=\frac{1}{a\alpha+b}$$

PSS

easy to get if $f(t) = e^{\alpha t}$

$\underbrace{A \times (\text{natural})}$ + particular is always a soln of

$$a\frac{dv}{dt} + bv = f(t)$$

plug it in & verify

LHS of DE $= A\underbrace{\left(a\frac{d(\text{natu})}{dt} + b\,\text{nat}\right)}_{\text{by def of natural respon} = 0} + \left(a\frac{d(\text{par})}{dt} + b\,\text{par}\right)$ by defn $= f(t)$ $(e^{\alpha t})$

$= A \cdot 0 +$ (par)

(Superposition principle of DE)

A => initial condns

(=> our circuit superposition)

Back to it all: we need soln for

$$RC\frac{dv}{dt} + v(t) = V\,\underline{\sin(\omega_0 t)}$$

$$\frac{e^{j\omega_0 t} - e^{j\omega_0 t}}{2j}$$

Could we est( ) particular soln

Guess particular soln for $V\sin(\omega_0 t)$

$V\sin(\omega_0 t + \phi)$

$RC\,V\cos(\omega_0 t + \phi)\,\omega_0 + V\sin(\omega_0 t + \phi) = V\sin(\omega_0 t + \phi)$

eqn in terms of ϕ = ??

If I calc/solve => ✓ But, takeaway

=> the form is correct

Do Laplace but => work

(as L^{-1} is messy)